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Thermodynamic analysis and experimental investigation of the water spray cooling of photovoltaic solar panels

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Abstract

This paper investigates an alternative cooling method for photovoltaic (PV) solar panels by using water spray. For the assessment of the cooling process, the experimental setup of water spray cooling of the PV panel was established at Sultanpur (India). This setup was tested in a geographical location with different climate conditions. It was found that the temperature of the panel decreased from 53 to 23 °C and the total power was increased by 15.3% by the water spray cooling. The effectiveness of the system is also increased by its cleaning effects. The efficiency of this solar PV is reduced with the increase of panel temperature. The experiments showed that the PV cell efficiency was dropped by 0.5% with an increase of 1 °C in panel temperature. However, the electrical efficiency of the panel was increased by 0.28%/0.2 °C of temperature drop by the single nozzle spray cooling.

Keywords Thermodynamic · Spray cooling · Photovoltaic · Solar · Water spray method · Efficiency

List of symbols

A	Area/m^{-2}
R_e	Reynolds number
PV	Photovoltaic
WBT	Wet bulb temperature/°C
DBT	Dry bulb temperature/°C
Q	Heat transfer rate/W
$\dot{m}$	Mass flow rate/kg s^{-1}
T	Temperature/K
C_p	Specific heat at constant pressure/kJ kg^{-1} K^{-1}
PCM	Phase change material
G	Solar irradiation/W m^{-1}
I	Current/A
V	Voltage/V
h	Heat transfer coefficient/W m^{-2} K^{-1}
N_u	Nusselt number
P_r	Prandtl number
K	Thermal conductivity/W m^{-2} K^{-1}
v	Velocity/m s^{-2}

Greek symbols

α	Absorptivity
ε	Emissivity
σ	Boltzmann constant/W m^{-2} K^{-4}
η	Efficiency/%
μ	Dynamic viscosity/N s m^{-2}

Subscripts

p	Panel
loss	Loss
C	Convection
E	Evaporation
R	Radiation
0	Ambient condition
w	Wall
f	Fluid

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Introduction

Nowadays, environmental protection and efforts to reduce pollution, caused by industrial activities, on the one hand, and research on finding new and improved energy supply options, on the other, have become one of the concerns of governments all around the world [1–3]. In recent years, photovoltaic (PV) power plants due to their proven potential in most parts of the earth are rapidly developing. In several studies, the negative effect of increasing the surface temperature of photovoltaic modules on their electrical efficiency has been shown. The temperature coefficient for power generation depends on the photovoltaic technology of the solar cells [4, 5]. According to the literature review, the highest test temperature for the photovoltaic module was 125 °C in Libya. This temperature caused a 69% drop in the electricity production of PV modules [6]. PV cells are extensively used as one of the most important renewable energy applications because they can use solar energy by converting solar irradiance to direct current (DC) power [7]. According to the materials used to make PV cells, solar irradiance can be converted into direct electricity with varying conversion efficiency ratings ranging from 7 to 40% [8]. A range of about 80% of solar radiation incidents on the surface of PV cells can be absorbed; however, due to the conversion efficiency of PV cell manufacturing technology, only a tiny percentage of the absorbed incident solar energy is transformed into electrical energy [9]. The remaining absorbed energy overheats the PV cells which can reach an operating temperature of around 40 °C over the ambient atmospheric temperature [10–12]. PV cells overheat because they convert a certain band of the entering irradiance spectrum wavelength that is responsible for light direct conversion into electrical energy, while the remaining spectral wavelength causes the PV cells to overheat [13]. Elevated PV panel temperatures are considered one of the most essential issues, especially in hot climatic locations, as they cause a sequential decline in PV electrical conversion efficiency of roughly 0.5%/°C for every degree increase in PV panel temperature, lowering PV panel lifetime [14]. Excess overheating needs to be drained in order to reach an acceptable level of PV cell electricity conversion efficiency because high temperature reduces the performance of the PV cells. Therefore, it is a vital responsibility for enhancing PV cell performance and assuring an efficient conversion process, especially in sun-belt nations. It is necessary to integrate the cooling system for PV cells during the operation. Furthermore, the cooling system will aid in lowering the overall cost of solar cells, extending the lifetime of PV cells, promoting solar cell manufacturing and ensuring optimum output power from installed PV cells [15].Water and air are the most common cooling methods. Air cooling requires less energy than that of water cooling. However, water has a greater cooling capacity and can be circulated for recooling. Also, warm water can be utilized for other purposes such as bathing and washing [16]. Active and passive cooling systems are the most common methods of lowering solar panel temperatures. Water and air are used as a coolant in active cooling, which is driven by a fan or an electric motor. Passive cooling can be further categorized as conductive passive cooling, water cooling and air passive cooling. Heat exchangers, sinks [17] and heat pipes are used to create natural convection in other passive cooling technologies. There are various cooling technologies such as phase change materials (PCM), nanotechnology (nanofluid) sinks, thermoelectric generators, microchannel and other modern cooling methods [18]. In this direction, Shalaby et al. [19] investigated solar panel water cooling is more efficient than air cooling. Water spray cooling enhances the total power output of photovoltaic panels by 33.3%. Spray cooling of water reduced the temperature by 57.1% from 24.7 to 26.4 °C. Also in [20], the authors investigated the effect of evaporative cooling implemented on PV panels and the maximal detected a total increase in power output was around 19%. Direct PV panel cooling with an established water flow over the front side of the panel was investigated in [21] and it was possible to increase power output by 9.5%. Furthermore, in [22] the authors investigated a water spray cooling technique implemented just on the front side of the PV panel, and significant improvement in electrical efficiency was established. Apart from water cooling, there are various other techniques to cool solar PV panels such as microchannel heat exchanger cooling [23], solar panel nanofluid cooling [24], solar panel evaporative cooling [25] and PCM cooling. In the PCM cooling method, latent heat and melting point temperature are the important parameters that decide the cooling rate. PCM absorbs or releases significant quantities when it is realized to change their physical state, and PCM has a high absorber capacity during phase change [26]. In this field recently, Foteinis et al. [27] have experimented with the PCM cooling method of PV panel cooling. They examined that by cooling with PCM the PV panel electricity efficiency was increased by 9.4%. Also, the environmental footprint is reduced by 22% compared to the standalone PV cell. In another cooling method, Lupu et al. [28] examined that heat pipe cooling improved the thermal efficiency of the PV panel by 13.9%. Apart from this, heat pipe uses sealed pipe which should have a high value of thermal conductivity like copper–silver, etc. The heat pipe converted solar panel heat to air or water; this lowered the system heat and improved the system efficiency. It was observed from the literature survey that various researches were present on solar panel

water spray cooling, but only a few manual works achieved above 25% efficiency. The novelty of the present work was to evaluate the performance of the PV panel with water spray cooling with a single nozzle and its thermal aspect for Sultanpur (India). The main objective of this paper was to develop an experimental setup and to investigate a water spray cooling technique for Sultanpur (India). The proposed water spray cooling technique can potentially increase PV panel performance due to an evaporation and self-cleaning effect, which is also a great benefit in terms of improved feasibility in the long run.

Experimental setup

The setup for an experiment was made to study the performance of a photovoltaic panel with spray cooling. The solar panel water spray cooling system remains on the roof of the hostel of KNIT Sultanpur, India, for several days during June 2022 on a typical clear summer day when average temperatures of the surrounding air ranged from 30 °C and up to 35 °C. Measurements were taken from 1 to 3 PM (the period of highest solar irradiation levels during June). During the series of measurements, irradiation ranged from 810 W m^{-2} to 850 W m^{-2}, in the already mentioned period of highest solar irradiation (the specific recorded peak value was 850 W m^{-2}) [29]. The geographical location of this place is the latitude, longitude and altitude, respectively, which are 26.27°N, 82.07°E and 95 m above sea level. The main components of this experiment are a solar panel nozzle strainer, a motor pump and a tank. The solar panel is located at some angle from the earth's surface. The panel was fixed under a specific angle of 18° (to obtain the highest electricity output for the fixed slope and which is characteristic of the specific geographical location where the panel was tested). It is situated on the four-pillar stand. Water is pulled from the tank by an electric motor which is dipped inside the reservoir (therefore, it is not visible in Fig. 1) and sprayed on the panel surface by the nozzle. The nozzles were on the front side of the panel and they were fixed at an angle of 40° (to prevent a wire frame shadowing effect and also to ensure a wider water spray dispersion over the PV panel as much as possible) and it did not analyze the nozzle angle influence on the panel performance. After spraying the cooled panel temperature, therefore, panel temperature falls from the rated temperature, and water fully slides on the panel surface so that with the cooling action, the cleaning action is also done. This water, again and again, recalculates via a softer pipe through the pump which is put inside the bucket. The mass flow of water is constant with the value of 2.3 kg s^{-1}. The water distribution system was mounted on a flexible pipe of diameter 4 mm. The distance between the nozzle and panel surface has been taken as 150 mm front side [30]. It was fixed for the maximum dispersion of the water on the panel. It was observed that as compared to the other cooling systems water spray cooling was more efficient, except

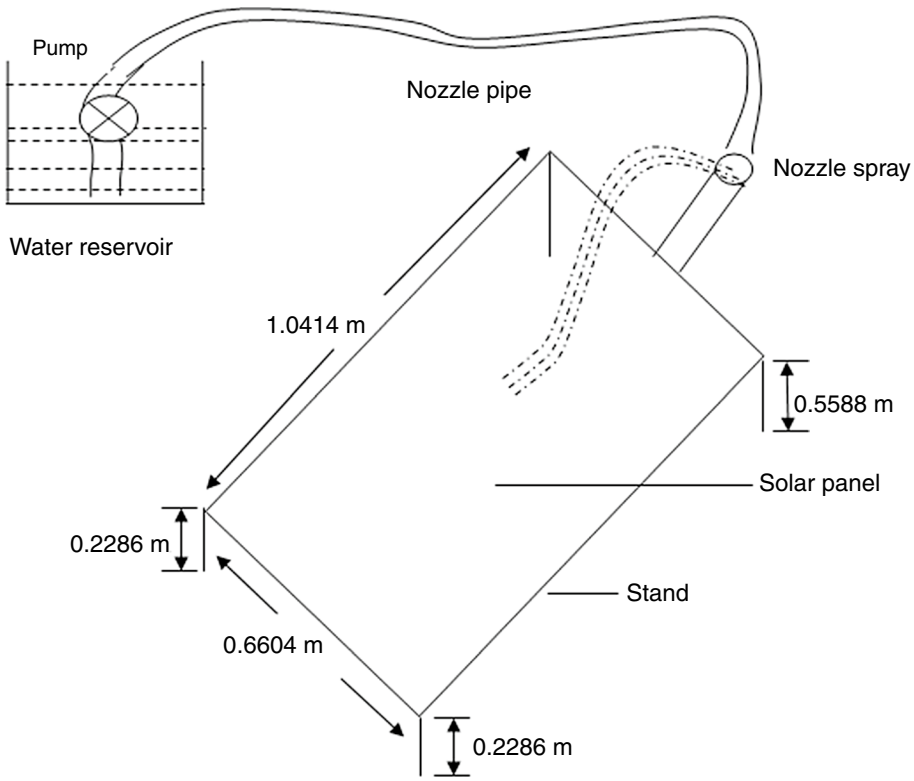


Fig. 1 Designed water spray cooling system for solar panels

for that water submerged method in which higher efficiency was obtained. Also, water spray works as a self-cleaning agent with a good cleaning action technique. The surface of the panel is made of a low coefficient of reflex ion material. Less reflection means more light is observed by the surface and also it has a lower mass than that of other materials. When there is solar irradiation incidence on the PV cells same time, infrared radiation was observed efficiently by the surface. The remaining light is converted into electricity by the photovoltaic cell. The setup of the line diagram and photograph is shown in Figs. 1 and 2, respectively.

The main components used in the solar panel water spray cooling system and their specifications are described as a *solar panel stand:* A stand is necessary for holding the panel. It has four legs. It is constructed of iron material and parts are welded having dimensions 1.0414 m × 0.6604 m. The lengths of the lower and higher legs are 0.2286 m and 0.5588 m, respectively. It has a nozzle mouth at the higher leg side. The nozzle diameter mouth is 1.51 mm as displayed in Fig. 3. *Nozzle pipe:* A PVC material pipe is used for transferring water from the bucket to the nozzle mouth. It has a length of 1.5 m and 1.38 mm in diameter as shown in Figs. 1 and 3. *Motor:* The motor is used for the conversion of mechanical energy into hydraulic energy. It generates flow with power and, according to the load, manages the pressure. An AC motor integrated pump 18 W, 165-230 V/50 Hz, model MEN265A (Mechanic Pvt. Limited) is used for pulling water from the reservoir through the pipe and send to the nozzle pipe. Its photograph is shown in Fig. 3. *Nozzle:* For spraying water, a 1.38 m diameter nozzle is used. It is made of steel material with many small holes to spray the water. *Solar panel:* Photovoltaic modules use the generation of electricity by the use of solar energy (photons). The modules utilized thin-film cells or crystalline silicon cells.

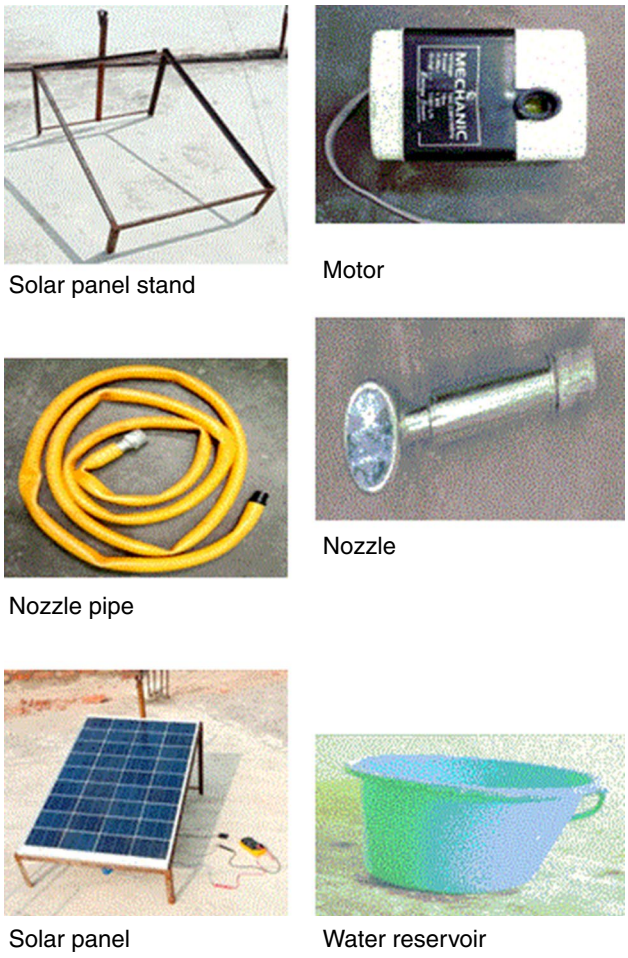


Fig. 3 Components for experimental setup

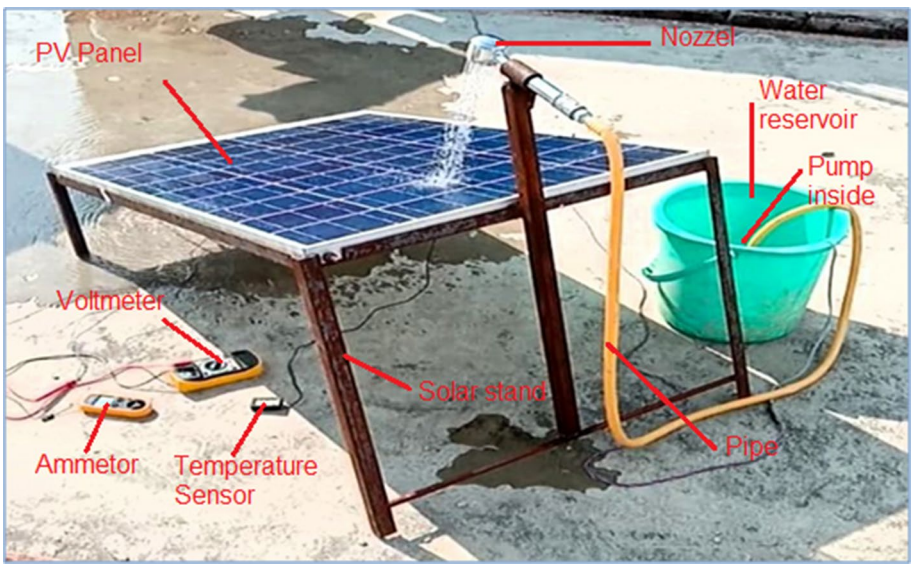


Fig. 2 Photograph of solar PV panel with water spray cooling system (experimental setup)

Table 1 Technical characteristics of solar PV panel

Characteristics*	
Model	LUM 12,165
Effective area/m^{-2}	0.6877
Maximum power output/W	165
Open-circuit voltage/maximal voltage/V	22.84 V/18.65 V
Short-circuit current/maximal current	9.55A/8.85A

*At solar irradiation (DNI) of 950 W m^{-2} and temperature of 25 ℃

It was developed by Luminous Technologies Pvt. Limited, India. The characteristics of the panel are given in Table 1. In all cells, one electric cell is so connected that mechanical damage and vibration do not take place. A 12 V power solar panel is used for cooling purposes. Its length is 1.0414 m and its width is 0.6604 m. This solar panel is fixed on the solar panel stand. The conversion of solar energy into electrical energy by the solar panel is given in Figs. 1 and 3. *Water reservoir:* A water reservoir capacity of 20 L is used for the flow of water. It is used for the continuous supply of water. The motor pump is fully dipped inside it and pulled water from it.

The following instruments are used for the measurements of the system characteristics. *Temperature sensor:* The temperature at the required places (backside and front side as shown in Fig. 1) was measured by the digital sensor and is shown in Fig. 4. This sensor (model TPM-10) can be used to measure the temperature in the range of -50 °C to 110 °C with an accuracy of + 1 °C. This temperature sensor is manufactured by ApTechDeals Company. *Whirling hygrometer:* Whirling hygrometer is used for the measurement of measuring wet and dry bulb temperatures. It can measure temperatures up to 20°F as shown in Fig. 4. *Anemometer:* The wind velocity affects the temperature of the solar panel; its speed is measured by an anemometer as shown in Fig. 4. The range of velocity of the anemometer is 0–45 m s^{-1} with a range of operating temperature – 10 to 50 °C and humidity 40–85% RH, respectively. *Voltmeter:* Voltmeter is used for measuring voltage and current value. It can measure the maximum voltage as 600 V, and the maximum current is 10A. It has three poles: One is voltage, another is current, and the third, between the two, is a neutral pole. *Electronic digital caliper:* Electronic digital caliper is used for measuring the dimension of different sections. It can be measured in units like mm and inch. It can measure maximum dimensions in the range of 150 mm. Its photograph is shown in Fig. 4.

In summer conditions, the temperature of the solar panel becomes very high; consequently, the value of power output is reduced. Therefore, it is necessary to cool the solar panel for better performance.

Measurement procedure and thermodynamics

The water is taken out from the water reservoir by the pump. This water goes through the pipe and is sprayed on the panel by the nozzle. The panel is cleaned and also cooled with this water. The reading was taken every 5–10-min intervals, and the values of voltage, current, dry bulb temperature (DBT) and wet bulb temperature (WBT) were noted. Also, the air velocity was noted because velocity affects the panel temperature due to forced convection.

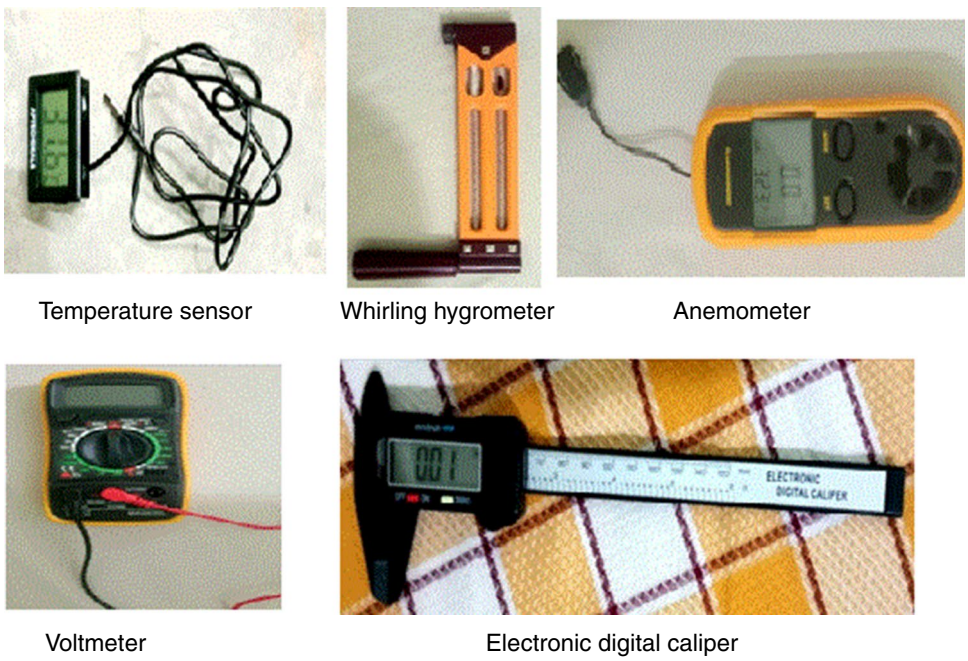


Fig. 4 Instrument used in the experimental setup

The water spray cooling technique aims to achieve a lower solar panel temperature in this way by increasing the output power of PV. Let G_S be the incoming solar irradiation and A_P be the total solar panel area. Now we are interested in converting the use of full power output of this incoming solar radiation.

According to this Fig. 5, Q_{loss} is the total heat loss, in convection, it is Q_C, Q_E is the total evaporation heat and Q_R is the radiation heat loss.

Then the total solar irradiation is expressed as [30, 31]:

$$Q_{solar} = \alpha \cdot G_S \cdot A_p \tag{1}$$

where α, G_S and A_p absorptive coefficient, solar irradiation and panel area, respectively.

Overall photovoltaic heat loss can be calculated by [30, 31]:

$$Q_C + Q_R + Q_E = Q_{loss} \tag{2}$$

where Q_E, Q_R and Q_C are evaporative, radiation and convection heat loss, respectively.

Total convection heat loss in front and backside is [30, 31]:

$$Q_{C,F} + Q_{C,B} = Q_C \tag{3}$$

Convection heat loss from the front side is also calculated by given below [30, 31]:

$$Q_{C,F} = h_{front} \cdot A_p \cdot \left(T_{panel\ front} - T_{air\ front}\right) \tag{4}$$

Same as from the backside [30, 31]:

$$Q_{C,B} = h_{backside} \cdot A_p \cdot \left(T_{panel\ backside} - T_{air\ backside}\right) \tag{5}$$

Total radiator heat loss is expressed as [30, 31]:

$$Q_{R,F} + Q_{R,B} = Q_R \tag{6}$$

Total heat radiation Q_R can also be calculated by [30, 31]:

$$Q_R = \sigma \cdot \varepsilon \cdot A_P \cdot F_{xy}\left(T_x^4 - T_y^4\right) \tag{7}$$

where x subscription denoted that front side any description denoted that backside, and $F_{xy,}$, T_x, σ, A_P and ε are the view factor of both side, front side temperature, backside temperature, Stefan–Boltzmann constant, panel area and emissivity, respectively.

The total evaporation heat loss depends on both relative humidity and surrounding air temperature and also the velocity of air, respectively. Total evaporation heat loss in PV panels can be explained as [30, 31]:

$$Q_E + Q_{EF} = Q_{E,B} \tag{8}$$

The evaporation heat loss also can be written as [30, 31]:

$$Q_E = e \cdot A_p \cdot \left(P_s - P_d\right) \cdot r \tag{9}$$

The purpose of the water spray is to increase overall heat reflection by evaporation, which is strongly influenced by the evaporation coefficient (e). P_s and P_d is the partial pressure, and the total evaporation heat loss depends on the relative humidity velocity of the air and the temperature of the water spray, where the evaporation coefficient greatly depends on the air velocity; in case of turbulent flow, it directly depends on the concoction heat transfer coefficient. Additionally, it relies on the temperature of narrow flow separation, relative humidity of the surrounding air and other factors.

Also, the following equations have been considered for the performance calculation. Hence, it is clear that heat transfers via force convection.

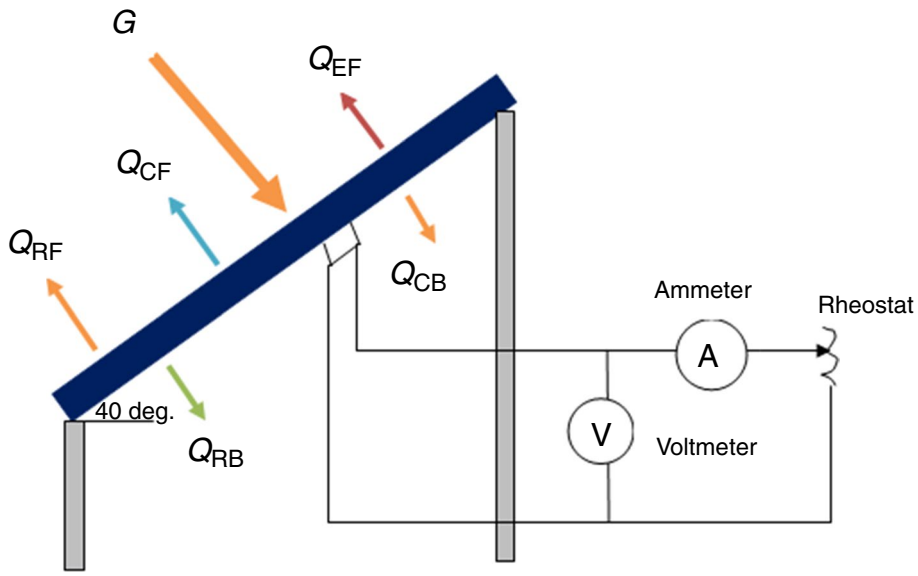


Fig. 5 Image of net heat loss from solar panel

According to Newton's cooling law [30, 31]:

$$Q = h \cdot A_s \cdot \left(T_w - T_f\right) \tag{10}$$

Here T_f and T_w are the temperature of fluid and wall, respectively. Q is related to overall heat transfer.

According to the second energy balance [30, 31]:

$$Q = \dot{m} \cdot C_p \cdot \left(T_e - T_0\right) \tag{11}$$

The flow is checked for laminar or turbulent flows by the Reynolds number [30, 31]:

$$R_e = \frac{\rho \cdot V \cdot d}{\mu} \tag{12}$$

where ρ, v and μ are density, velocity and dynamic viscosity, respectively,

The velocity of fluid can be determined by [30, 31]:

$$\dot{m} = \rho \cdot A \cdot V \tag{13}$$

Nusselt number can be calculated as [30]:

$$N_u = 0.332\left(R_e\right)^{1/2}\left(P_r\right)^{1/3} \text{ (laminar)} \tag{14}$$

$$N_u = 0.0288\left(R_e\right)^{4/5}\left(P_r\right)^{1/3} \text{ (turbulent)} \tag{15}$$

The formula for the heat transfer coefficient (h) is [30, 31]:

$$N_u = \frac{hd}{k} \tag{16}$$

where d is the hydraulic length.

The efficiency of the system can be calculated as [30, 31]:

$$= \frac{\text{Output power}}{A_p \cdot G_s}. \tag{17}$$

Results and discussions

In this section, performance of the system was discussed without cooling and with cooling one after the other. The results of the experiment were recorded without cooling of solar panel in the month of July 10, 2022, as shown listed in Table 1. The experiment analysis has been taken during the time 1:00 PM to 3:00 PM a day. In this period, there is maximum solar irradiation incidence on the PV panel [29]. This solar irradiation is direct normal irradiation (DNI) (W m.$^{-2}$) incidence on the PV panel. The wind velocity, DBT, WBT and module power have been recorded at some stage in the test and were used within the calculation of the PV module efficiency. The panel temperature initially at 60.6 °C (1:00 PM) is increased and reached the peak point of the day at 62.3 °C (2:00 PM). During the experiment, the panel temperature fluctuated, according to wind velocity and environmental conditions. The minimum temperature of the panel is 59.8 °C at 2:30 PM. After this time, the temperature becomes down gradually. It can be observed that both the value of voltage as well as the value of current drop with increasing cell temperature. It has been noted that as temperatures rise, the module efficiency marginally declines. The solar panel's initial temperature at 1:00 PM was 60.6 °C, and its efficiency was 2.88%. When the temperature slightly decreases to 60.1 °C, then efficiency slightly improved to 2.919%. All variation are listed in Table 2

Further in this section, all measurements were taken with the cooling of the solar panel on July 12, 2022, at Sultanpur, India. The recorded data are listed in Table 3. The wind velocity, DBT, WBT and module power were recorded at some stage in the test and used within the calculation of the PV module efficiency. The panel temperature initially at 45 °C (1:00 PM) is lowered and reaches the lowest point of the day at 42.8 °C (3:15 PM). During the experiment, the panel temperature fluctuated, according to wind velocity and environmental conditions. The minimum temperature of the panel is 42.8 °C at 3:15 PM. After cooling the temperature become down gradually. Here, it was observed that both the value of voltage and the quantity of current result

Table 2 Recorded data of solar panel without cooling on July 10, 2022

Time	1:00 PM	1:15 PM	1:30 PM	1:45 PM	2:00 PM	2:45 PM	3:00 PM
Temperature/°C	60.6	60.1	60.8	61.2	62.3	61.8	59.8
Voltage/V	19.08	19.30	19.24	19.11	19	19.03	19.28
Current/A	1.04	1.06	0.98	0.93	0.63	0.58	0.08
DBT/°C	89	89.1	90	90	91	89.9	88
WBT/°C)	82.5	82.4	83	83	83.5	03	82.9
Velocity of air/ms^{-1}	1.09	1.2	1.1	1.3	1.08	1.2	1.08
Output power/W	19.84	20.458	18.85	17.77	12.92	11.03	20.822
Efficiency/%	2.88	2.919	2.69	2.536	1.84	2.33	3.027

Table 3 Recorded data of solar panel with cooling on July 12, 2022

Time	1:00 PM	1:15 PM	1:30 PM	1:45 PM	2:00 PM	2:15 PM	2:30 PM	3:00 PM	3:15 PM
Temperature/°C	45.0	44.9	44.6	44.5	44.3	46.3	46.3	43	42.8
Voltage/V	20.6	21.9	21.2	20.9	21.9	19.6	20.6	22.1	22.3
Current/A	4.54	4.75	4.96	4.84	4.90	4.75	4.58	4.80	4.83
DBT/°C	80	79.9	79.8	79.7	79.8	80	80	79.9	79.9
WBT/°C	74	73.9	73.8	73.8	73.7	74	74	73.9	73.9
Velocity of air/m s^{-1}	1.1	1.09	1.33	1.08	1.45	1.06	1.3	1.09	1.8
Output power/W	93.5	100.7	105.152	101.156	107.31	91.14	95.264	106.08	107.709
Efficiency/%	13.5	14.64	15.29	14.709	15.60	13.25	13.856	15.42	15.662

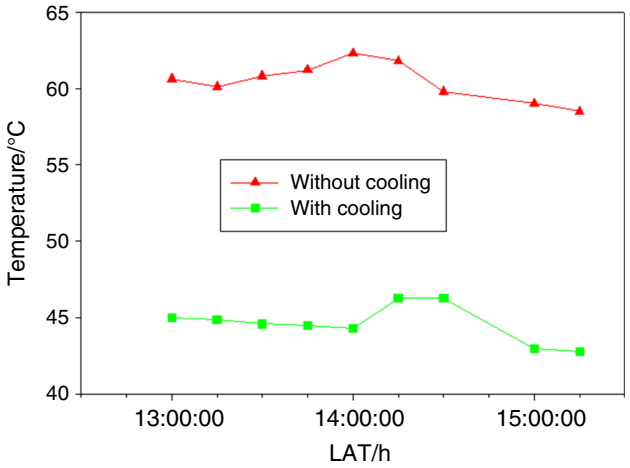


Fig. 6 Variation of temperature with time

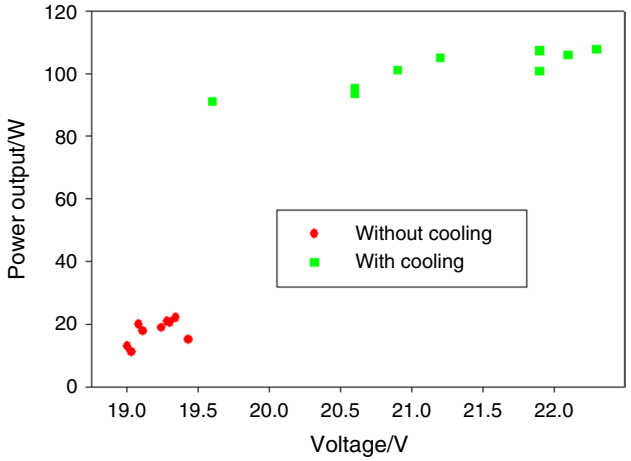


Fig.7 Power output variation with voltage

in decreasing cell temperature. Additionally, it is evident from the results of the experiment that the module efficiency steadily raises with temperature. At 1:00 PM, it is found that the initial temperature of a solar panel is 45 °C and the efficiency is 13.5%. When the temperature slightly decreases to 42.8 °C, then efficiency is slightly improved to 15.662%. At 1:15 PM, the efficiency is 14.64%; after the 15 min, the temperature becomes 44.6 °C which was lowered by 0.2 °C and the efficiency was improved to 15.2% from 14.64%. Therefore, the solar panel must be cooled for achieving better performance and efficiency.

Variation of temperature

Figure 6 shows the temperature variation of solar panel with time. It is observed that the maximum temperature is between 1:00 PM and 2:30 PM on July 10, 2022. At this time, a high amount of solar radiation was received in India. Therefore, energy transformation rate was also high. It can be observed that solar panel temperature fluctuated with the environmental condition and wind velocity also. Further it also shows the temperature variation with time with cooling. The maximum temperature was observed as 46.23 °C in the whole test range, i.e., between 1:00 PM and 3:15 PM on July 12, 2022. The minimum temperature was found as 42.8 °C at 3:15 PM while in the previous test, i.e., without cooling, it was 59.75 °C at the same time. Apart from this, panel temperature fluctuated with the environmental condition and wind velocity as observed in Fig. 6. Solar irradiation is the critical parameter for any solar-integrated study. The variation of surface temperature with direct normal irradiation (DNI) has been also discussed. There is no need to show the variation of the temperature with the DNI because it will show the same trend as the temperature with time, because the same trend showed DNI with the temperature [29].

Variation of power output with voltage

The maximum power was achieved at the time 3:00 PM when the minimum temperature was found. The maximum power was obtained as 20.822 W at 59.8 °C temperature. Figure 7 shows that power output decreased as the voltage increased, reaching a maximum of 19.28 V. It is concluded that when the panel temperature tends to maximum then

their power output becomes minimum. Also, in this subsection variations of electrical power with voltage with cooling were discussed. The maximum was observed at 3:15 PM when the temperature was the lowest of all-temperature readings. Table 3 shows that at 42.8 °C maximum voltage and the maximum power were found as a maximum of 22.3 V and 107.709 W. It is concluded that when the panel temperature tends to maximize, the corresponding power would be minimized. The comparison showed that due to the cooling effect, the efficiency was improved. As discussed above, the electrical efficiency decreased with panel surface temperature. It means if it is cooled then it will improve. Figure 7 shows that the dispersion points without cooling are very nearly distributed. It means the variation of the power is very less with the voltage produced. It is because of the high temperature of the panel. Without cooling, power varied only 19.5–19 V during the same tested time and solar irradiation. However, with cooling the dispersion point on the graph is widely sprayed. It means due to the spray of the water over the panel temperature and cleaning is there. The variation of the power with voltage varied from 90.3 to 107.7W as the wide range of voltage varied from 19 to 22.5 V.

Variation of current with voltage

It was also observed that the current first increased with the voltage, and after some value of the voltage, it decreased. Keeping constant all other parameters, as the voltage increased the corresponding temperature also decreased which leads to the decrease in the current. The maximum current was found at 1.06 V of voltage. Figure 8 shows that the dropping rate of the current was fast corresponding to the voltage from 0.58 to 0.55 V. As discussed previously, for this reason the rate of temperature drop was also sharp. Further current variation with the voltage with cooling was also discussed. The pattern of variation was the same as that without cooling conditions. It was observed that the maximum current value occurred at the highest value of the voltage. As the panel temperature lowered, both temperature and voltage got the highest value. In comparison with Fig. 8, it was seen that with cooling the current was obtained more than without cooling. It can be seen that little bit of variation of the current with voltage without cooling the panel. Without cooling, the current was varied from 0.18 to 1.2A only because the temperature of the panel was high before cooling the duct and other impurities are there. However, a high current was achieved with the cooling, and the current varied from 4.8 to 4.9A in between fluctuating with the voltage. This variation occurred with a wide range of voltage, i.e., 19.0 to 22.1 V.

Variation of efficiency with temperature

It was observed that when the panel temperature reached a maximum value the corresponding electrical efficiency also obtained its minimum value as shown in Fig. 9. In this experiment, the maximum efficiency occurred at the lowest temperature which is 59.8 °C, and at the maximum temperature 60.8 °C, efficiency is the lowest value which is 2.69%. Electrical efficiency first increased slowly with temperature; after that, it increased sharply from the temperature of 46.3–43 °C. Maximum electrical efficiency was found as 22.32% at 43 °C overall temperature limits during the test with cooling, while lowest, at temperature 46.3 °C, efficiency has a value of 13.20% as displayed in Fig. 9. The comparison showed that the panel's electrical efficiency was more in cooling as compared to without cooling. This is due to a decrease in temperature and the cleaning action of water. The variation of the efficiency

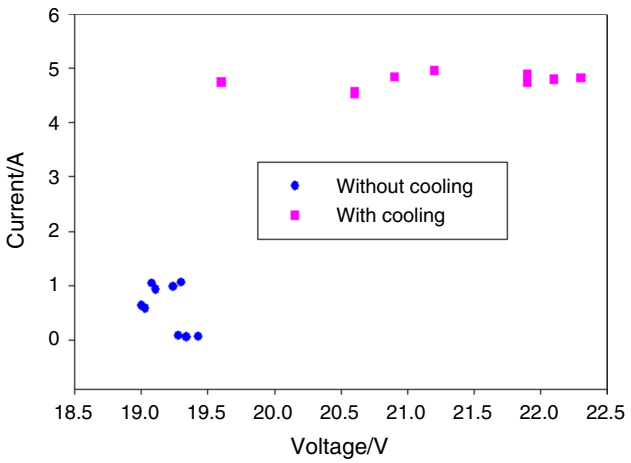


Fig. 8 Variation of current with voltage

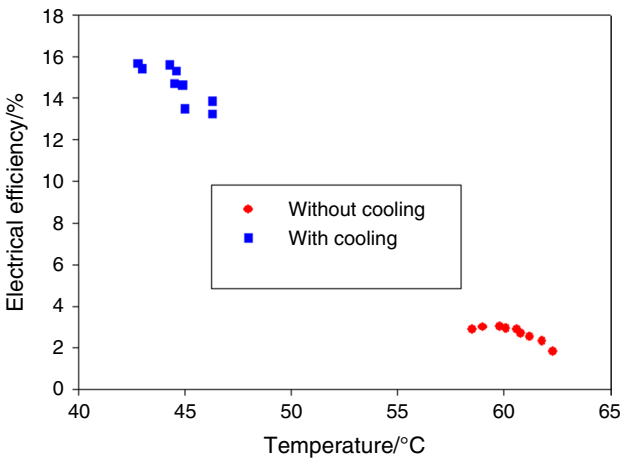


Fig. 9 Efficiency variations with temperature

with the temperature without cooling varied from 2.88 to 3.027% when the temperature increased from 60.6 to 59.8 °C during the tested time, i.e., 1:00 PM to 3:00 PM. However, the efficiency will increase with cooling with the temperature from 13.5 to 15.667% when the temperature increased from 45.0 to 42.8 °C. The temperature increased with time with cooling the efficiency also decreased.

Performance variation with temperature

The power of the module increases; hence, its efficiency increases. Higher temperatures of solar panels reduce both power and efficiency. It can be maintained by cooling the heated solar panel. Figure 10 shows the variation of power and efficiency with temperature. The efficiency is found below 20% and current power fluctuated between 80 and 110 W, while maximum power output and efficiency are obtained at 42.8 °C which is the minimum temperature in the experiment. The range of solar irradiation is 810–850 W m^{-2}. The power output will increase from 80 to 110W with the temperature from 42.8 to 46 °C. The temperature fluctuates. The highest temperature was found as 46 °C at which the efficiency and power were 13.25% and 91.5%, respectively. The highest electrical efficiency was found to be 110 kW at the 42.2 °C surface temperature.

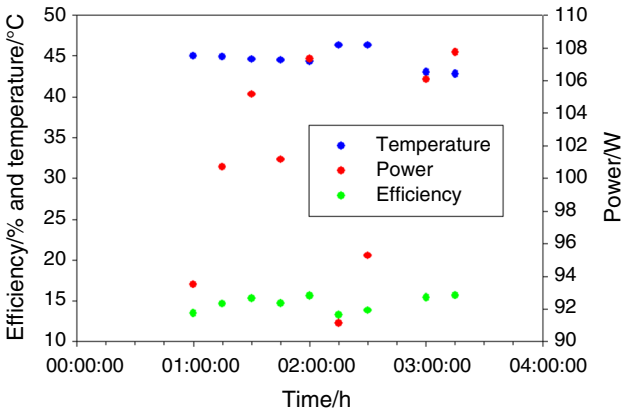


Fig. 10 Variation of power and efficiency with temperature

Comparisons of the result with previous work

The performance of the solar panel water cooling (July 2022) is compared with the previous spray water cooling system for photovoltaic panel work (June 2019) [31]. The compassion is made by considering the parameter of solar panel temperature, solar radiation and the solar panel system efficiency in both cooling and non-cooling systems. While the time frame for the current project was 1:00 PM to 3:15 PM at 15-min intervals, the prior project's time frame was 11:30 AM to 3:30 PM at 10-min intervals. The time and date for the current experiment are different from the previous study. In earlier research, the temperature range was between 24.2 and 27.8 °C with a spray cooling system. However, in the present research, the temperature range was between 60.8 and 59 °C with a spray cooling system. The narrowing of the temperature range is brought on by wind speed, water's particular heat, DBT and WBT. The results are listed in Table 4. In the reference, the panel's maximum power output was obtained values of 54W and 72W without and with spray cooling, respectively. In reference research work, where the highest power production is 22.822W without a cooling panel, employing the steady cooling system results in increases in maximum electrical power output of roughly 33.3%. This maximum power is increased to 107.7 W by continuous spray cooling in the present work as given in Table 4. The present research with the single point pray cooling increased the power output by 80.66%. In the case without a cooling system, electrical efficiency was found to mean an efficiency of 9.11%. Utilizing the stable cooling system, the prior work's electrical efficiency was obtained at 12.1%. The present work maximum efficiency without cooling was obtained as 3.027%. This efficiency improved by cooling to 15.662%. The maximum voltage value in the previous research work was recorded at the voltage of 13.5 V with the cooling system this is improved to 17 V, whereas in current research work 19.23 V in the case without a cooling system, this is improved to 22.3 V with spray cooling.

Table 4 Comparison of the results with previous studies under the same operating conditions*

Performance parameters	Reference [31]		This study	
	Without cooling	With cooling	Without cooling	With cooling
Electrical efficiency/%	9.1	12.1	3.02	15.66
Power output/W	54	72	20.82	107.7
Voltage/V	13.5	17	19.23	22.3

*Average DNI of 985 W m^{-2} and ambient temperature of 28–31 °C

Conclusions

The thermodynamic analysis is performed on the experimental setup of the water spray cooling of photovoltaic solar panels. The following main conclusions can be obtained from this study:

- The highest electrical efficiency and power of the PV panel were found as 15.66% and 107.8 W, respectively, at the minimum temperature of 42.8 °C at the time of 3:15 PM.
- The maximum temperature of the solar panel is found between 40 and 60 °C. Higher temperature prompt lowered the efficiency and power output of the solar panel.
- The electrical efficiency of the panel was increased by 0.28%/0.2 °C of temperature drop by the single nozzle spray cooling.
- The environment, wind speed, dry bulb and wet bulb temperatures, panel inclination, water temperatures, and distance between nozzle and solar panel influenced the efficiency and performance of the solar panels.
- By water spray cooling, many types of dust particles (an environmental impurity that is laid on the panel surface) are cleaned, and consequently, higher solar irradiation is achieved. Consequently, thermal performance was improved.

Author contributions YK and RR had written the complete paper which was conceptualized by MMR and HC. Moreover, the proofreading was carried out by MMR and HC. All the experiments and data collection were performed by MKC and AKC under the esteemed guidance.

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